to their visual capabilities, making the app more perceivable. The three modes are - Deuteranomaly (red-green), Protanomaly (red-green) and Tritanomaly (blue-yellow).

Colour inversion and colour correction changes are not reflected in screenshots. Hence, a picture of the phone's screen was taken in order to show the changes. Figure 12 shows the effect of colour inversion; Figures 13 and 14 show the effect of colour correction. Figure 13 shows the screen before selecting the option, and Figure 14 shows the screen after selecting Tritanomaly.

Tips for developers

Google has its own checklist for developers to help them make their applications accessible, some of which are listed below:

1. Provide content descriptions (the *android: contentDescription* attribute should be used in the layout file) for elements such as images. These behave like alternate text descriptions for images.
2. Ensure correct navigation. Users may navigate through the app using other hardware such as an external keyboard. It is the responsibility of the developer that the focus is shifted properly when the user presses the tab key, or the arrow keys.
3. Audio feedback should be supported by captions or vibration.
4. Test the accessibility of the app using TalkBack and other assistive technologies, such as an external keyboard connected to the device.

In addition, developers should take care of the following aspects too:

1. The user should be provided with options to change the text size, text colour, background colour, etc, within the app.
2. The size of the elements on the screen should be appropriate. There should be proper spacing between two elements so that users feel free to move around with the help of their fingers when using a touchscreen device.

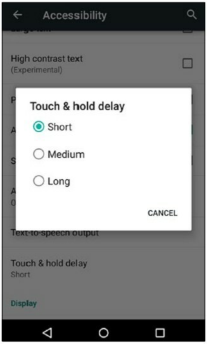

Figure 11: Setting touch and hold delay

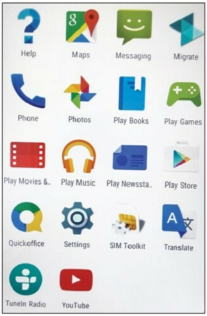

Figure 13: Before selecting colour correction option

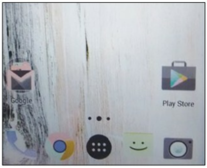

Figure 12: Effect of colour inversion

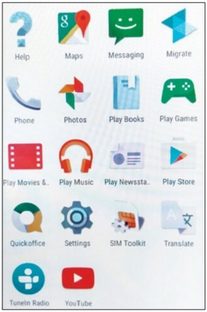

Figure 14: Colour correction after selecting Tritanomaly

3. Test your app on various versions of Android.
4. Interact with people with impairments and understand the problems they face while they use mobile apps. Also, let them test the app that you have developed. This will help identify the flaws in the design as well as gauge how well the app functions. END

References

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Interested readers may also read the author's paper on the accessibility of popular applications on Google Play at *http://www.indjst.org/index.php/indjst/article/view/61436.*

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